

Example

The compression ratio of an ideal Otto cycle is 8. At the onset of the compression stroke, the pressure is 0.1 MPa & the temperature is 15°C. The heat supplied to air, per cycle, is 1800 kJ/cycle. Determine:

1. The pressure & the temp. for each state.
2. The thermal efficiency → max. efficiency you can get.
3. The mean effective pressure. $MEP = \frac{W_{net}}{V_{max} - V_{min}}$

Solution

Given

$r = 8$
 $P_1 = 0.1 \text{ MPa}$
 $T_1 = 15^\circ\text{C}$
 $Q_{in} = 1800 \text{ kJ/cycle}$

Required

1. $P_2, T_2, P_3, T_3, P_4, T_4$
2. η_{th}
3. MEP

Assumptions

1. Air standard assumptions are applicable.
2. KE & PE are negligible.
3. The variation of specific heats with temperature is to be neglected.

Analysis

- start with simple p-v (state) (with many givens)

→ state 1 - state 2 (1 → 2)

$$\left. \begin{matrix} P_1 = 0.1 \text{ MPa} \\ T_1 = 288 \text{ K} \end{matrix} \right\} \Rightarrow V_1 = \frac{RT_1}{P_1} = 0.822 \text{ m}^3/\text{kg}$$

What is the process b/n 1 & 2

→ Isentropic

→ Ideal gas

→ pick isentropic ideal gas relation/formula

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = r^{\gamma-1} = 8^{\gamma-1} = 8^{1.4-1} = 8^{0.4} = 2.297$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = r^{\gamma} = 8^{\gamma} = 8^{1.4} = 18.98 \Rightarrow P_2 = 0.1 \text{ MPa} \times 18.98 = 1.898 \text{ MPa}$$

$$V_2 = \frac{V_1}{r} = \frac{0.822 \text{ m}^3/\text{kg}}{8} = 0.1027 \text{ m}^3/\text{kg}$$

Process 3-4

Isentropic expansion

Isentropic relation to be used

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1} = r^{\gamma-1}$$

$$T_4 = 1380 \text{ K}$$

$$\frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^{\gamma} = r^{\gamma}$$

$$P_4 = 0.479 \text{ MPa}$$

Process 2-3

Isentropic process heat addition

→ $V = C$: what relations to use? Energy balance & ideal gas eqn

Energy balance: 1st law THD for closed system

$$(Q_{in} - Q_{out}) + (W_{out} - W_{in}) = \Delta U$$

$$Q_{in} + (W_{out} + W_{in}) = \Delta U$$

$$Q_{in} + \int P dv = \Delta U: V = C$$

State 3

$$Q_{23} = U_3 - U_2 = C_v(T_3 - T_2)$$

$$T_3 = T_2 + \frac{Q_{23}}{C_v} = 3174 \text{ K}$$

$$\Rightarrow PV = RT$$

$$P_3 V_3 = RT_3 \text{ \& } P_2 V_2 = RT_2 \text{ but } V_3 = V_2$$

$$\frac{P_3}{P_2} = \frac{T_3}{T_2} \Rightarrow P_3 = \frac{T_3}{T_2} P_2 = 8.813 \text{ MPa}$$

$$C_v = f(T) = \left(\frac{\partial U}{\partial T}\right)_v \Rightarrow \Delta U = U_3 - U_2 = \int_{T_2}^{T_3} C_v(T) dT$$

Data: are needed $T_{ref} = 0 \text{ K}$

$$\eta_{th} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{8^{1.4-1}} = 0.565$$

$$\rightarrow \eta_{th} = 56.5\%$$

$$Q_{out} = C_v(T_4 - T_3) = 782.3 \text{ kJ/kg}$$

Example

The compression ratio of an ideal Otto cycle is 8. At the onset of the compression stroke, the pressure is 0.1 MPa & the temperature is 15°C. The heat supplied to air, per cycle, is 1800 kJ/rp. Determine:

1. The pressure & the temp. for each state.
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Solution

Given

$r = 8$
 $P_1 = 0.1 \text{ MPa}$
 $T_1 = 15^\circ\text{C}$
 $\dot{Q}_{in} = 1800 \text{ kJ/rp}$

Required

1. $P_2, T_2, P_3, T_3, P_4, T_4$
2. η_{th}
3. MEP

Assumptions

1. Air standard assumptions are applicable.
2. KE & PE are negligible.
3. The variation of specific heats with temperature is to be neglected.

Analysis

- start with simple p-t (state) (with many givens)

→ state 1 - state 2 (1 → 2)

$$P_1 = 0.1 \text{ MPa} \Rightarrow V_1 = \frac{RT_1}{P_1} = 0.827 \text{ m}^3/\text{rp}$$

$$T_1 = 288 \text{ K}$$

What is the process b/n 1 & 2

↳ Isentropic

↳ ideal gas

↳ pick isentropic ideal gas relation/formula

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = r^{\gamma-1} = 8^{1.4-1} = 8^{0.4} = 2.297396 \Rightarrow T_2 = 288 \times 2.297396 = 661.65 \text{ K}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = r^{\gamma} = 8^{1.4} = 18.97 \Rightarrow P_2 = 0.1 \text{ MPa} \times 18.97 = 1.897 \text{ MPa}$$

$$\frac{V_2}{V_1} = \frac{V_2}{r} = \frac{0.827}{8} = 0.1034 \text{ m}^3/\text{rp}$$

process 3-4

Isentropic expansion

Isentropic relation to be used

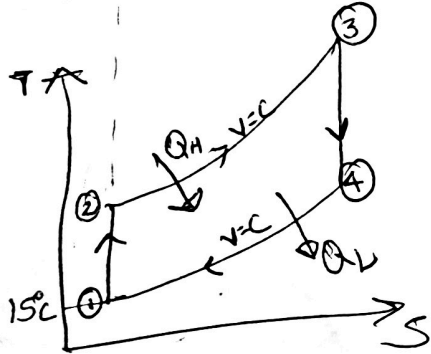
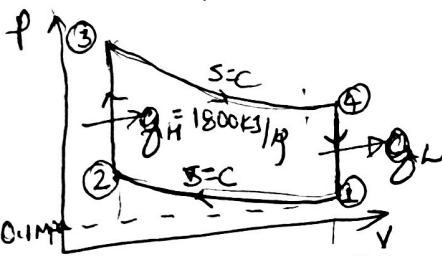
$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1} = r^{\gamma-1}$$

$$T_4 = 1380 \text{ K}$$

$$\frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^{\gamma} = r^{\gamma}$$

$$P_4 = 0.479 \text{ MPa}$$

$$\dot{Q}_{out} = (C_v(T_4 - T_1)) = 782.3 \text{ kJ/rp}$$



Process 2-3

Isobaric process heat addition

↳ $V = C$: what relations to use? Energy balance ideal gas C_v

Energy balance: 1st law THD for closed system

$$(\dot{Q}_{in} - \dot{Q}_{out}) + (W_{out} - W_{in}) = \Delta U + \Delta KE + \Delta PE$$

$$\dot{Q}_{in} + (W_{out} - W_{in}) = \Delta U$$

$$\dot{Q}_{in} + \int P dV = \Delta U : V = C$$

$$\text{State 3 } \dot{Q}_{23} = U_3 - U_2 = C_v(T_3 - T_2)$$

$$T_3 = T_2 + \frac{\dot{Q}_{23}}{C_v} = 3174 \text{ K}$$

$$\Rightarrow PV = RT$$

$$P_3 V_3 = RT_3 \text{ \& } P_2 V_2 = RT_2 \text{ but } V_3 = V_2$$

$$\frac{P_3}{P_2} = \frac{T_3}{T_2} \Rightarrow P_3 = \frac{T_3 P_2}{T_2} = 8.813 \text{ MPa}$$

$$C_v = f(T) = \left(\frac{\partial U}{\partial T}\right)_V \Rightarrow \Delta U = U_3 - U_2 = \int_2^3 C_v(T) dT$$

Values are ready $T_{ref} = 0 \text{ K}$

$$2. \eta_{th} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{8^{1.4-1}} = 0.565$$

$$\Rightarrow \eta_{th} = 56.5\%$$

$$\eta_{th} = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}}$$

Example:

A Diesel cycle operates on air with a low pressure of 0.1 MPa & a low temperature of 15°C. If the compression ratio is 16 & the heat supplied is 1800 kJ/kg. Compute:

1. The pressure, the temperature & the specific volume for each state.
2. The cutoff ratio.
3. The thermal efficiency.
4. The mean effective pressure.

Analysis

Process 1→2 / Isentropic compression (s=c)

State 1: $T_1 = 15^\circ\text{C} = 288\text{ K}$
 $P_1 = 0.1\text{ MPa} = 100\text{ kPa}$

($PV = RT$) $V_1 = \frac{RT_1}{P_1} = 0.827\text{ m}^3/\text{kg}$

State 2: Isentropic, Ideal gas situation

$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} = r^{k-1} = 16^{k-1}$

$T_2 = 873.5\text{ K}$

$\Rightarrow \frac{P_2}{P_1} \left(\frac{V_1}{V_2}\right)^k = r^k = 16^k$

$P_2 = 4.85\text{ MPa}$

$\Rightarrow \frac{V_1}{V_2} = r \Rightarrow V_2 = \frac{V_1}{r} = 0.0517\text{ m}^3/\text{kg}$

Process 2→3 Isobaric process (p=c)

State 3: $P_3 = P_2 = 4.85\text{ MPa}$

1st law of THD for process 2→3: $\sum E_{in} - \sum E_{out} = \Delta E_{system}$
 $(\dot{Q}_{in} + \dot{W}_{in}) - (\dot{Q}_{out} + \dot{W}_{out}) = \Delta U + \Delta KE + \Delta PE$

$\dot{Q}_{in} = C_p(T_3 - T_2) = 1800\text{ kJ/kg}$

$\Rightarrow T_3 = \frac{\dot{Q}_{in}}{C_p} + T_2 = \frac{1800}{1.0035} + 873.5\text{ K}$

$T_3 = 2667\text{ K}$

$V_3 = ?$ considering $\frac{P_3}{T_3} = \frac{P_2}{T_2} = \frac{RT_2}{V_2}$

$PV = RT$
 $P = \frac{RT}{V}$

Solution

Given

Diesel cycle
 $Air: k = 1.4$
 $P_1 = 0.1\text{ MPa}$
 $T_1 = 15^\circ\text{C}$
 $r = 16$

$\dot{Q}_{in} = 1800\text{ kJ/kg}$

Required

1. (P, T, V) for each state

2. r_c

3. $\eta_{th, Diesel}$

4. MEP

Assumptions

- The cold air standard assumptions are applicable, thus air can be assumed to have constant sp. heats at room temperature.

- ΔKE & ΔPE are negligible.

$C_p =$

$C_v =$

@ 288 K C_p

250 - 1.003 0.716

288 - ? ?

300 - 1.005 0.718

$\frac{38}{50} = \frac{288 - 250}{300 - 250} = \frac{C_p - 1.003}{1.005 - 1.003} = \frac{C_v - 0.716}{0.718 - 0.716}$
 $C_p = 1.00452$

$\frac{V_3}{V_2} = \frac{T_3}{T_2}$

$\Rightarrow V_3 = 0.1578\text{ m}^3/\text{kg} : V_3 > V_2$

$\Rightarrow$ The piston moved down in heat addition
 Process 3→4 Isentropic expansion (s=c)

↳ Ideal gas & Isentropic relations

$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{k-1} = \left(\frac{0.827}{0.1578}\right)^{k-1}$ | $V_1 = V_4$
 $\frac{T_3}{T_4} = \left(\frac{V_1}{V_2}\right)^{k-1} = \left(\frac{0.827}{0.0517}\right)^{k-1}$ | $= 0.827\text{ m}^3/\text{kg}$

↳ Not r like Otto

$\Rightarrow T_4 = 1375\text{ K}$ $\Rightarrow$ still high; it can be recovered before rejection to produce? Φ electric power for car!...

$\Rightarrow \frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^k$

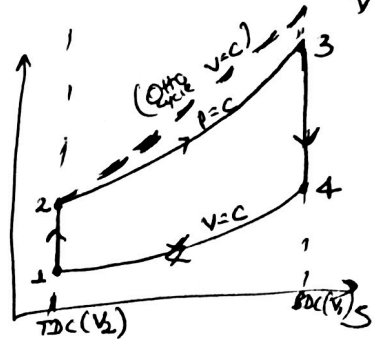
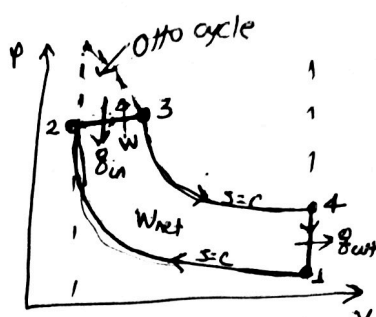
$\Rightarrow P_4 = 0.048\text{ MPa}$

$V_4 = V_1 = 0.827\text{ m}^3/\text{kg}$

2. r_c (cutoff ratio)

$r_c = \frac{V_3}{V_2} = \frac{0.1578}{0.0517} = 3.05$

3. $\eta_{th, Diesel} = \frac{W_{net}}{\dot{Q}_{in}} = 1 - \frac{\dot{Q}_{out}}{\dot{Q}_{in}} = 1 - \frac{C_v(T_4 - T_1)}{C_p(T_3 - T_1)} = 1 - \frac{0.716(1375 - 288)}{1.00452(2667 - 288)} = 0.567 = 56.7\%$
 and $\dot{Q}_{out} = C_v(T_4 - T_1) = 0.7165(1375 - 288) = 778.3\text{ kJ/kg}$



MEP = Average pressure that will give you the same W_{net} used to compare reciprocating engines

The target of Thermo II is not the actual efficiency but (Otto $\eta = 20-25\%$)
 → Maximum efficiency we can reach

100%
 ↓
 Carnot η
 ↓
 The system η

Example

The compression ratio of an ideal Otto cycle is 8.

At the onset of the compression stroke, the pressure is 0.1 Mpa & the temp is 15°C.

The heat supplied to air, per cycle, is 1800 kJ/kg. Determine

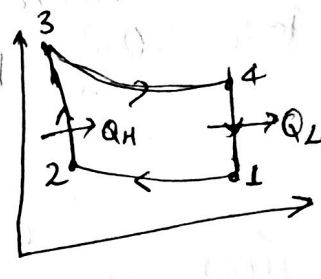
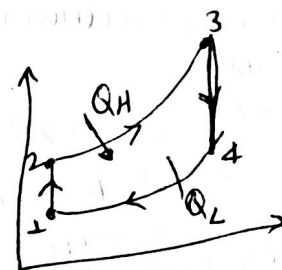
1. The p & T for each state
2. The thermal efficiency
3. The mean effective pressure

Assumption

Given

Reqd

Assumptions



Analysis

- Get all temp & pressures: start from the simple

State 1

$$P = 0.1 \text{ Mpa}$$

$$T_1 = 288 \text{ (15+273)}$$

$$PV = RT$$

$$\Rightarrow V_1 = \frac{RT_1}{P_1}$$

$$= 0.827 \text{ m}^3/\text{kg}$$

State 2 Isentropic

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} = r^{\gamma-1} = 8^{\gamma-1}$$

$$T_2 = 662 \text{ K}$$

$$\Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = r^{\gamma} = 8^{\gamma}$$

$$P_2 = 1.838 \text{ Mpa}$$

$$V_2 = \frac{V_1}{r} = 0.1034 \text{ m}^3/\text{kg}$$

State 3 Isochoric

$$Q_{in} = Q_{23} = C_v (T_3 - T_2)$$

$$T_3 = T_2 + \frac{Q_{in}}{C_v} = 3174 \text{ K}$$

State 4

$$V_3 = V_2, \text{ so } P_3 = T_3 \frac{P_2}{T_2} = 8.98 \text{ Mpa}$$

$$\frac{P_3 T_3}{P_2 T_2} = \frac{R T_3}{R T_2}$$

State 4

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1} = r^{\gamma-1} = 8^{\gamma-1}$$

$$V_3 = V_2, V_1 = V_2$$

$$T_4 = 1380 \text{ K}$$

$$\frac{P_3}{P_4} = \left(\frac{V_4}{V_3}\right)^{\gamma} = r^{\gamma} = 8^{\gamma}$$

$$\Rightarrow P_4 = 0.479 \text{ Mpa}$$

$$\Rightarrow \eta_{th} = 1 - \frac{1}{r^{\gamma-1}} = 0.565$$

$$\eta = 56.5\%$$

$$Q_{out} = -C_v (T_1 - T_4) = 780.3 \text{ kJ/kg}$$

since $Q_{in} = Q_{out}$